WORKED SOLUTION

STEP III 2021, Question 4

Pure • Vectors and Projections

STEP III 2021, Question 4

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Everything here is done with the scalar product: an angle between two vectors is read off from $\mathbf{u}\cdot\mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos(\text{angle})$, and projection is linear, so the projection of $\frac{1}{2}(\mathbf{a}+\mathbf{b})$ is $\frac{1}{2}(\mathbf{a}_1+\mathbf{b}_1)$. Part (i) shows that the average of two *equal-length* vectors bisects the angle between them. In part (iii) the projections $\mathbf{a}_1$ and $\mathbf{b}_1$ usually have different lengths, so we should expect bisection to fail unless something special happens. The condition factorises neatly into the two special cases.

Solution

Part (i)

Since $|\mathbf{a}| = |\mathbf{b}| = 1$ and $\mathbf{a}\cdot\mathbf{b} = \cos\theta$,

$$|\mathbf{m}|^2 = \tfrac{1}{4}\left(1 + 2\cos\theta + 1\right) = \tfrac{1}{2}(1+\cos\theta) = \cos^2\tfrac{\theta}{2},$$

so $|\mathbf{m}| = \cos\frac{\theta}{2}$, which is positive because $0 < \theta < \pi$. In particular $\mathbf{m} \neq \mathbf{0}$. Also

$$\mathbf{m}\cdot\mathbf{a} = \tfrac{1}{2}(1+\cos\theta) = \cos^2\tfrac{\theta}{2} = \mathbf{m}\cdot\mathbf{b}.$$

So the cosine of the angle between $\mathbf{m}$ and $\mathbf{a}$ is $\cos^2\frac{\theta}{2}\big/\cos\frac{\theta}{2} = \cos\frac{\theta}{2}$, and the same holds for $\mathbf{b}$. Both angles are therefore $\frac{\theta}{2}$. Since $\mathbf{m}$ lies in the plane of $\mathbf{a}$ and $\mathbf{b}$ and the two angles add up to θ, $\mathbf{m}$ bisects the angle between $\mathbf{a}$ and $\mathbf{b}$.

Part (ii)

By definition, with $\mathbf{c}$ as the unit normal and $\mathbf{a}\cdot\mathbf{c} = \cos\alpha$,

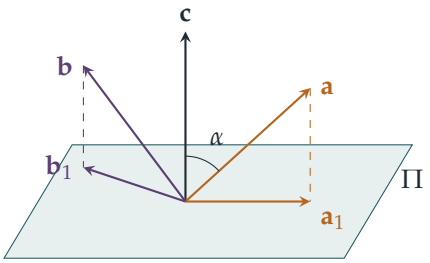
$$\mathbf{a}_1 = \mathbf{a} - \cos\alpha\,\mathbf{c}, \qquad \mathbf{b}_1 = \mathbf{b} - \cos\beta\,\mathbf{c}.$$

Then $\mathbf{a}_1\cdot\mathbf{c} = \cos\alpha - \cos\alpha\,(\mathbf{c}\cdot\mathbf{c}) = 0$, since $|\mathbf{c}| = 1$. Similarly $\mathbf{b}_1\cdot\mathbf{c} = 0$.

Because $\mathbf{a}_1 \cdot \mathbf{c} = 0$, we get $|\mathbf{a}_1|^2 = \mathbf{a}_1 \cdot (\mathbf{a} - \cos\alpha\,\mathbf{c}) = \mathbf{a}_1 \cdot \mathbf{a}$, and

$$\mathbf{a}_1 \cdot \mathbf{a} = \mathbf{a} \cdot \mathbf{a} - \cos\alpha\,(\mathbf{c} \cdot \mathbf{a}) = 1 - \cos^2\alpha = \sin^2\alpha.$$

As α is acute, $\sin\alpha > 0$, so $|\mathbf{a}_1| = \sin\alpha$. In the same way $|\mathbf{b}_1| = \sin\beta$. Both are nonzero, so the angle ϕ is defined.



For the angle between the projections, again use $\mathbf{a}_1 \cdot \mathbf{c} = 0$ to drop a term:

$$\mathbf{a}_1 \cdot \mathbf{b}_1 = \mathbf{a}_1 \cdot (\mathbf{b} - \cos\beta\,\mathbf{c}) = \mathbf{a}_1 \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{b} - \cos\alpha\,(\mathbf{c} \cdot \mathbf{b}) = \cos\theta - \cos\alpha\cos\beta.$$

Dividing by $|\mathbf{a}_1||\mathbf{b}_1| = \sin\alpha\sin\beta$ gives

$$\cos\phi = \frac{\cos\theta - \cos\alpha\cos\beta}{\sin\alpha\sin\beta},$$

as required.

Part (iii)

Projection is linear, so

$$\mathbf{m}_1 = \mathbf{m} - (\mathbf{m} \cdot \mathbf{c})\,\mathbf{c} = \tfrac{1}{2}(\mathbf{a} - \cos\alpha\,\mathbf{c}) + \tfrac{1}{2}(\mathbf{b} - \cos\beta\,\mathbf{c}) = \tfrac{1}{2}(\mathbf{a}_1 + \mathbf{b}_1).$$

We assume $\mathbf{m}_1 \neq \mathbf{0}$, since otherwise it has no direction and cannot bisect anything. As $\mathbf{m}_1$ is a positive combination of $\mathbf{a}_1$ and $\mathbf{b}_1$, it lies in their plane and between them. So it bisects the angle exactly when it makes equal angles with $\mathbf{a}_1$ and $\mathbf{b}_1$:

$$\frac{\mathbf{m}_1 \cdot \mathbf{a}_1}{|\mathbf{a}_1|} = \frac{\mathbf{m}_1 \cdot \mathbf{b}_1}{|\mathbf{b}_1|}.$$

(We have cancelled the common factor $|\mathbf{m}_1|$, and cos is one-to-one on $[0, \pi]$.)

? **Ask yourself** Why not just apply part (i) to $\mathbf{a}_1$ and $\mathbf{b}_1$?

Part (i) needed $|\mathbf{a}| = |\mathbf{b}|$. Here $|\mathbf{a}_1| = \sin\alpha$ and $|\mathbf{b}_1| = \sin\beta$, which are usually different. So we must compute.

Write $P = \mathbf{a}_1 \cdot \mathbf{b}_1$, $s = \sin\alpha$ and $t = \sin\beta$. Then

$$\mathbf{m}_1 \cdot \mathbf{a}_1 = \tfrac{1}{2}(s^2 + P), \qquad \mathbf{m}_1 \cdot \mathbf{b}_1 = \tfrac{1}{2}(P + t^2).$$

Multiplying through by $2st > 0$, the condition becomes

$$\begin{aligned} t(s^2+P) = s(t^2+P) &\iff st(s-t) - P(s-t) = 0 \\ &\iff (s-t)(st-P) = 0. \end{aligned}$$

So $\mathbf{m}_1$ bisects the angle if and only if $s = t$ or $P = st$.

- $s = t$ means $\sin\alpha = \sin\beta$. Since sin is strictly increasing on $(0, \frac{\pi}{2})$ and both angles are acute, this holds if and only if $\alpha = \beta$.
- By part (ii), $P = \cos\theta - \cos\alpha\cos\beta$. So $P = st$ means

$$\cos\theta = \cos\alpha\cos\beta + \sin\alpha\sin\beta = \cos(\alpha - \beta).$$

Each step was an equivalence, so $\mathbf{m}_1$ bisects the angle between $\mathbf{a}_1$ and $\mathbf{b}_1$ if and only if $\alpha = \beta$ or $\cos\theta = \cos(\alpha - \beta)$.

Geometrically, the first case says $\mathbf{a}_1$ and $\mathbf{b}_1$ have equal length, so part (i) applies after all. The second says $\cos\phi = 1$: the projections point the same way (when $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are coplanar with $\mathbf{a}$ and $\mathbf{b}$ on the same side of $\mathbf{c}$). The angle is then zero, and it is bisected trivially.

Checks

- Take $\alpha = \beta = \frac{\pi}{2}$ (not allowed here, but a useful limit). Then $\mathbf{a}$ and $\mathbf{b}$ already lie in the plane, and the formula gives $\cos\phi = \cos\theta$, as it should.
- With $\alpha = \frac{\pi}{3}$, $\beta = \frac{\pi}{4}$ and $\theta = \frac{\pi}{2}$, the formula gives $\cos\phi = -\frac{\sqrt{2}/4}{\sqrt{6}/4} = -1/\sqrt{3} \approx -0.577$. This agrees with constructing the vectors explicitly.
- Both the formula for $\cos\phi$ and the factorised bisection condition were confirmed by computer on several thousand random triples of unit vectors.

Common slips

- Forgetting that $|\mathbf{a}_1| \neq 1$. The projection of a unit vector is not a unit vector, so the denominator $\sin\alpha\sin\beta$ is essential.
- Assuming the bisector of $\mathbf{a}_1, \mathbf{b}_1$ is always $\mathbf{a}_1 + \mathbf{b}_1$. That holds only for equal lengths. In general the bisector is $\mathbf{a}_1/|\mathbf{a}_1| + \mathbf{b}_1/|\mathbf{b}_1|$.
- Dividing by $\sin\alpha - \sin\beta$ instead of factorising, and so losing the case $\alpha = \beta$.

Beyond STEP

Projection onto a plane is a *linear map* $\mathbf{x} \mapsto \mathbf{x} - (\mathbf{x}\cdot\mathbf{n})\mathbf{n}$, with matrix $I - \mathbf{n}\mathbf{n}^{\mathsf{T}}$. Linear maps preserve sums but not lengths or angles, and that is why this question works out as it does. The formula for $\cos\phi$ is the spherical cosine rule in disguise. Here ϕ is the

angle at the vertex **c** of the spherical triangle with vertices **a**, **b** and **c**, whose sides are α, β and θ. For projections and orthogonality, see Strang, *Introduction to Linear Algebra*.

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